

Worksheet W10: Sine and Cosine

"The Circle Logic — The language of rotation and repetition."

Chapter 10 · 10 Problems · Name: _____ Date: _____

Derivative of $\sin(x) = \cos(x)$ | Derivative of $\cos(x) = -\sin(x)$

$\sin \rightarrow \cos \rightarrow -\sin \rightarrow -\cos \rightarrow \sin$

Problem 1

Question: What do **sine** and **cosine** measure on a unit circle?

💡 *Hint: One measures height, the other measures width.*

Problem 2

Question: What is the value of $\sin(0^\circ)$ and $\cos(0^\circ)$?

💡 *Hint: At 0° , the point is at the rightmost edge of the circle: $(1, 0)$.*

Problem 3

Question: What is the value of $\sin(90^\circ)$ and $\cos(90^\circ)$?

💡 *Hint: At 90° , the point is at the top of the circle: $(0, 1)$.*

Problem 4

Question: What is the **derivative** of $\sin(x)$?

💡 *Hint: The derivative of sine is cosine.*

Problem 5

Question: What is the **derivative** of $\cos(x)$?

💡 *Hint: The derivative of cosine is negative sine.*

Problem 6

Question: What is the **derivative cycle** for sine and cosine?

💡 *Hint: $\sin \rightarrow \cos \rightarrow -\sin \rightarrow -\cos \rightarrow \text{back to } \sin$.*

Problem 7

Find the derivative of $f(x) = \sin(2x)$.

💡 *Hint: Use the Chain Rule: derivative of $\sin(kx)$ is $k\cos(kx)$. Here $k = 2$.*

Problem 8

Find the derivative of $f(x) = \cos(3x)$.

💡 *Hint: derivative of $\cos(kx)$ is $-k\sin(kx)$. Here $k = 3$.*

Problem 9

Question: Find the integral of $\cos(x)$.

💡 *Hint: The integral of cosine is sine. Don't forget $+C$.*

Problem 10

Question: Why do sine and cosine repeat every 360° ?

💡 *Hint: Think about going around a circle — after one full rotation, you're back where you started.*